

TouchWall

For an interesting video on what touch technology should ideally be, head over to <http://www.perceptivepixel.com/> and watch Jeff Han manipulate a touch wall

Ocean waves

To get a first hand taste of Direct Compute check out this video that reproduces the physics of incredible looking wave crests – www.thinkdigit.com/d/ocean/

Feature

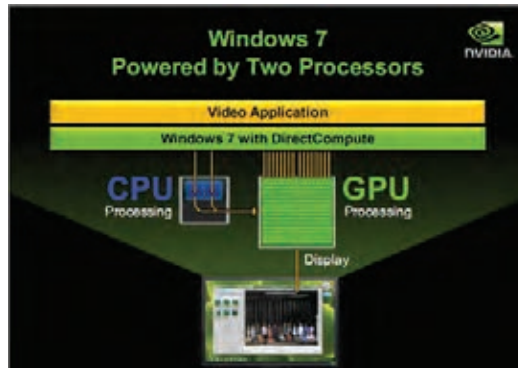
system. We were speaking earlier about the iPhone and how it's beautifully achieved the status of the numero uno touch device even today. This is because the device was built with touch in mind. Thought went into things like contextual menus and design elements like big buttons etc. Windows on the other hand has always been a mouse and keyboard interface. Now multi-touch is simply being ported to the OS.

According to Popli, the two platforms simply can't be compared. "Yes, for the iPhone the whole device was itself built around touch functionality. But one is a phone while here we're talking of a PC operating system. We have a billion users and we're not expecting a billion users to move over to touch functionality over night. Not every PC has to incorporate touch. But touch can and will greatly enhance the user experience," he said.

So basically all your basic mouse functions like clicking or scrolling are mapped to the touch interface. But the mouse is surely more precise at doing most things than fat old butterfingers, especially on an operating system that was designed for the pointer.

There are a few gestures that work universally throughout the operating system, which applications get automatically as a result of mapping mouse gestures or presence of scroll bars which are mapped to the mouse wheel. These include two finger scrolling or zooming via pinch gestures (ala iPhone), and right click via pressing and holding a tap, in addition to some programmable ones for common actions such as Copy and Undo.

But for applications to use touch to the fullest, touch functionality needs to be specifically incorporated via the SDK. How smartly they do it is entirely up to the application makers, since they too will have a fair bit of reinventing to do. So for now there's no immediate getting rid of the little rodent problem just as yet. Other issues like the hand obscuring the view in the table top form factor, greasy screens, lack of tactile response to make up for the loss of the cursor, will also come into the picture.



Windows 7 assigns operations to multiple processing elements

Touch applications

Still, there are some applications that wait out for a multi-touch interface but in the correct form factor. Maps and Google Earth is one example. Browsers are another area where touch can be a lot of fun. Imagine lying on the bed with a tablet and browsing Facebook pictures or YouTube videos. For whatever little typing, you can always bring up an on-screen keyboard. Touch may even greatly improve functionality of graphics and image editing suites like Photoshop or Illustrator. Alexander Phillip, a 3D modelling expert from



"Low memory consumption is very important, to drag icons or windows around the screen"

Igor Stanek, Product Manager, ION/Notebook, NVIDIA

Mumbai on the other hand says that he'd much rather use a mouse for precision or a Wacom Tablet. "Besides, why do away with those thousands of keyboard short cuts that you've grown used to over the years?" he adds.

Consider the gaming scenario. Can you imagine the kind of micro you will be able to perform while playing an RTS? The inference we can draw here is that touch support for application will not do away with conventional forms of input. With Windows 7, you will be able to use either touch, mouse, or keyboard as and when each makes sense.

"We've been asked if we are creating a new Touch UI, or "Touch Shell" for Windows – something like Media Center that completely replaces the UI of Windows with a version that is optimised for touch. We are focused on bringing touch through the Windows experience and delivering optimised touch interface where appropriate," says the Windows 7 Touch Team on the official blog.

Future

Other applications of this technology can be envisioned too. "What you can do with it is entirely up to what uses it can be put to through the API. Photo manipulation, making collages with drag and drop are just some of the things that you can do. Applications are being built for portfolio management, others include a newspaper reading application that would allow you to easily flip between pages with intuitive gestures," says Popli

Touch applications can be pictured virtually anywhere. "You can pick an industry and see how touch applications can be relevant within those industries." "In the Automobile segment I believe we're partnering with some of the companies here for applications leveraging touch functionality. Let's take for example if you happen to go to a showroom you can see a car and the exterior, experience a different look and feel. It will help you manipulate those customisations and data points in a much better way without going through the learning curve." In medicine teaching applications can be envisioned. You might be able to see a heart and touch a ventricle apply pressure to simulate a blockage at that point and the programme can respond.

With mobility comes miniaturisation over a period of time. "You'll see smaller laptops or netbooks and even tablets. Each will appeal to a different type of user," says Popli to round up. But to truly have such tablet-like portable computing devices to operate on a full touch environment, we believe there is no escaping a separate touch shell for Windows.¹